

# FYP DEVELOPMENT PLAN

Smart Academic Communication Platform – Development Plan



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# Table of Contents

1. Project Specifications ..... 2
1.1 High-Level System Components ..... 2
1.2 Documentation Structure ..... 2
1.2.1 Front Matter ..... 2
1.2.2 Chapters and Sections ..... 2
2. Team and Roles Distribution ..... 5
3. Development Timeline Strategy ..... 5
4. Deployment Strategy ..... 5

# 1. Project Specifications

## 1.1 High-Level System Components

In our final year project, there are following high-level components:

1. Authentication and Authorization
2. Academic Data Management
3. Template Management System
4. Campaign Management
5. Automation Rule Engine
6. Messaging Service and Notification System
7. Real-Time Communication Module
8. Email Classification and Prioritization System
9. Analytics and Reporting
10. File and Document Sharing Module

## 1.2 Documentation Structure

For the creation of the documentation of our project, I've created the following structure:

### 1.2.1 Front Matter

1. Cover Page
2. Statement of Submission
3. Acknowledgment
4. Abstract
5. Table of Contents
6. List of Figures
7. List of Tables

### 1.2.2 Chapters and Sections

1. Introduction
   1.1. Background
   1.2. Problem Statement
   1.3. Objectives
   1.4. Scope
   1.5. Methodology
   1.6. Technologies
2. Literature Review / Related Work
   2.1. Existing Tools comparison

3. System Analysis
    3.1. Requirement Gathering
    3.2. Functional Requirements
    3.3. Non-functional Requirements
    3.4. GANTT Chart
    3.5. Use-case Diagrams
    3.6. User Roles
    3.7. Domain Model Diagrams (DMDs)
    3.8. Data Flow Diagrams (DFDs)
4. System Design
    4.1. System Architecture
        4.1.1. Distributed System Architecture
        4.1.2. Service Communication Architecture
        4.1.3. Frontend Architecture
        4.1.4. Backend Architecture
        4.1.5. AI Architecture
        4.1.6. Automation Rule Engine Design
        4.1.7. Security Architecture
    4.2. Database Design
    4.3. Entity Relation Diagrams
    4.4. Interface Designs
    4.5. Component Designs
    4.6. Class Diagram
    4.7. Object Diagram
    4.8. Sequence Diagram
5. Implementation
    5.1. Development Tools and Environment
    5.2. Modules/Components Implemented
    5.3. Frontend Implementation
    5.4. Backend Implementation
    5.5. AI Service Implementation
    5.6. Database Implementation
    5.7. Authentication and Authorization
    5.8. Rule Engine Implementation
    5.9. Campaign Management Implementation
    5.10. AI Classification Implementation
    5.11. Realtime Messaging Implementation
6. AI/ML Methodology

- 6.1. Dataset Selection
- 6.2. Preprocessing
- 6.3. Feature Extraction
- 6.4. Training Process
- 6.5. Evaluation Metrics
- 6.6. Inference Workflow

7. Deployment Architecture
   - 7.1. Deployment Topology
   - 7.2. Hosting Strategy
   - 7.3. Cloud Architecture
   - 7.4. Environment Separation
   - 7.5. CI/CD Overview
8. System Testing
   - 8.1. Testing Techniques
     - 8.1.1. Unit Testing
     - 8.1.2. API Testing
     - 8.1.3. Realtime Communication
     - 8.1.4. AI Model Evaluation
     - 8.1.5. Security Testing
     - 8.1.6. Performance Testing
   - 8.2. Test Plans
   - 8.3. Test Cases
   - 8.4. Results and Bugs Reports
9. Results and Discussion
   - 9.1. Performance Metrics
   - 9.2. Evaluation of Results
     - 9.2.1. Screenshots
     - 9.2.2. Analytics
     - 9.2.3. AI Results
     - 9.2.4. Workflow Results
10. Project Management and Workflow
    - 10.1. Git Strategy
    - 10.2. Branch Flow
    - 10.3. Collaboration Workflow
    - 10.4. Development Methodology
    - 10.5. Sprint/Task Management
11. Conclusion and Future Work
12. References

## 2. Team and Roles Distribution

| TEAM MEMBER | ROLE |
|---|---|
| Talha Ahmad | Backend (Core) and Lead Developer |
| Muhammad Raza | Frontend Developer |
| Areeb Siddiqui | AI/ML Developer |
| Furqan Moin | Frontend and Real-Time Services Developer |

**Note:** Documentation creation role is shared among all team members depending on their contributions to the project's development.

## 3. Development Timeline Strategy

| SEMESTER | COMPONENT | DOCUMENTATION |
|---|---|---|
| **6** | 1 (Auth), 2 (Academic Data), 6 (Message Service) | 1, 2, 5 (1, 7), 10 (1, 2) |
| **7** | 3 (Template), 4 (Campaign), 8 (Classifier Model), 9 (Analytics) | 4 (1 [1, 2, 3, 4, 5, 7]), 5 (9, 10), 6 |
| **8** | 5 (Rule Engine), 7 (Real-Time Communication), 10 (File Sharing) | 4 (1 [6]), 5 (8, 11), 8, 9, 11 |
| ***SHARED AS PER REQUIREMENT*** | 1 (Auth), 6 (Message Service), 9 (Analytics) | 3, 4 (2, 3, 4, 5, 6, 7, 8), 5 (2, 3, 4, 5, 7), 7, 10 (3, 4, 5), 12 |

## 4. Deployment Strategy

| Layer | Hosting Platforms |
|---|---|
| Frontend (React) | Vercel / Netlify / Azure Static Web Apps |
| Backend Core (ASP.NET) | Azure App Service / Render |
| Realtime Services (Node.js) | Azure / Render |
| AI Services (Python) | Azure / Render |
| Database (Postgre SQL / Azure SQL) | Azure / Neon / Supabase |
| Redis | Upstash Redis |
| File Storage | Azure Blob Storage / Cloudinary |